

NCERT Exemplar Solutions For Class 12

Physics

Chapter 12-Atoms

Multiple Choice Questions

12.01 Taking the Bohr radius as $a_0 = 53\text{pm}$, the radius of Li^{++} ion in its ground state, on the basis of Bohr's model, will be about

- (a) 53pm
- (b) 27pm
- (c) 18pm
- (d) 13pm

Ans: (c): The radius of any ion in its ground state is given as $r = \frac{a_0}{Z}$ where a_0 is the Bohr's radius and Z is the atomic number. The Bohr's radius is equal to 53pm and the atomic number of lithium is three. So, the radius of Li^{++} ion in the ground state is $r = \frac{a_0}{Z} = \frac{53}{3} \text{pm} = 17.67 \text{pm} \approx 18 \text{pm}$. So, option (c) is the correct answer.

12.2 The binding energy of an H-atom, considering an electron moving around a fixed nuclei (proton), is $B = -\frac{me^4}{8n^2\epsilon_0^2h^2}$ (m=mass of electron) If one decides to work in a frame of reference where the electron is at rest, the proton would be moving around it. By similar arguments, the binding energy would be

$$B = -\frac{Me^4}{8n^2\epsilon_0^2h^2} \quad (M = \text{proton mass})$$

This last expression is not correct because

- (a) n would not be integral.

(b) Bohr - quantization applies only to electron

(c) The frame in which the electron is at rest is not inertial.

(d) The motion of the proton would not be in circular orbits, even approximately.

Ans: (c): Electron keeps on moving continuously in the orbitals. So, the frame in which electron is at rest cannot be inertial. Also, the electrostatic force cannot

provide the centripetal force $F_p = \frac{m_e v^2}{r}$ as the mass of electron is very less

(negligible) as compared to proton. So, option (c) is the correct answer.

12.3 The simple Bohr model cannot be directly applied to calculate the energy levels of an atom with many electrons. This is because

(a) Of the electrons, not being subject to a central force.

(b) Of the electrons colliding with each other

(c) Of screening effects

(d) The force between the nucleus and an electron will no longer be given by Coulomb's law.

Ans: (a): The simple Bohr model cannot be directly applied to calculate the energy levels of multi-electron system because the electronic repulsions between the electrons will interfere and the net force will not be subject to the central force. Hence, option (a) is the correct answer.

12.4 For the ground state, the electron in the H-atom has an angular momentum = $\frac{h}{2\pi}$, according to the simple Bohr model. Angular momentum is a vector and hence there will be infinitely many orbits with the vector pointing in all possible directions. In actuality, this is not true,

(a) Because Bohr's model gives incorrect values of angular momentum.

(b) Because only one of these would have a minimum energy.

(c) Angular momentum must be in the direction of spin of electron.

(d) Because electrons go around only in horizontal orbits.

Ans: (a): Angular momentum of a revolving electron is an integral multiple of $\frac{h}{2\pi}$ according to the Bohr's second postulate. Hence, the Bohr's model does not give correct value of angular momentum. So, option (a) is the correct answer.

12.5 O₂ molecule consists of two oxygen atoms. In the molecule, nuclear force between the nuclei of the two atoms

(a) is not important because nuclear forces are short-ranged.

(b) is as important as electrostatic force for binding the two atoms.

(c) Cancels the repulsive electrostatic force between the nuclei.

(d) is not important because oxygen nucleus have equal number of neutrons and protons.

Ans: (a): The nuclear forces are short ranged. Since the distance between the two oxygen atoms is larger as compared to the electrons of the two atoms. Hence, nuclear force between the nuclei of the two atoms is not important because nuclear forces are short-ranged. So, option (a) is the correct answer.

12.6 Two H atoms in the ground state collide in elastically. The maximum amount by which their combined kinetic energy is reduced is

(a) 10.20 eV

(b) 20.40 eV

(c) 13.6 eV

(d) 27.2 eV

Ans: (a): First we calculate the total energy of the two H atoms in the ground state.

So, total energy of two H-atom in ground state = $2(-13.6) = -27.2 \text{ eV}$

The maximum amount by which their combined kinetic energy is reduced when any one hydrogen atom goes into first excited state after the inelastic collision i.e., the total energy of two hydrogen atoms after inelastic collision is:

$$E = \frac{13.6}{n^2} + 13.6$$

Now, for n equals two. We have,

$$E = \frac{13.6}{2^2} + 13.6$$

$$\Rightarrow E = 3.4 + 13.6 = 17eV$$

So, Loss in kinetic energy due to inelastic collision = $27.2 - 17.0 = 10.2eV$

12.7 A set of atoms in an excited state decays.

- (a) in general to any of the states with lower energy.
- (b) Into a lower state only when excited by an external electric field.
- (c) All together simultaneously into a lower state.
- (d) To emit photons only when they collide.

Ans: (a): A set of atoms in an excited state decays in general to any of the states with lower energy.

Multiple Choice Questions-II More than one answer option

12.8 An ionized H-molecule consists of an electron and two protons. The protons are separated by a small distance of the order of angstrom. In the ground state,

- (a) The electron would not move in circular orbits.
- (b) The energy would be $(2)^4$ times that of an H-atom.
- (c) The electrons, orbit would go around the protons.
- (d) The molecule will soon decay in a proton and an H-atom.

Ans: (a) and (c): The nucleus of the two hydrogen atoms are separated by a small distance which is of order of angstrom. Also, the nuclear forces are of the same order. So, one electron will revolve around the protons. Hence, options (a) and (c) are the correct answers.

12.9 Consider aiming a beam of free electrons towards free protons. When they scatter, an electron and a proton cannot combine to produce an H-atom,

(a) Because of energy conservation.

(b) Without simultaneously releasing energy in the form of radiation.

(c) Because of momentum conservation.

(d) Because of angular momentum conservation.

Ans: (a) and (b): larger amount of energy is required for electron to reach near the proton as the binding energy of the hydrogen atom is larger as compared to the energy of a free electron. So, due to the energy conservation principle, it is not possible for electron and proton to combine and form hydrogen atom. Hence, options (a) and (b) are the correct answers.

12.10 The Bohr model for the spectra of an H-atom

(a) Will not be applicable to hydrogen in the molecular form.

(b) Will not be applicable as it is for a He-atom.

(c) is valid only at room temperature.

(d) Predicts continuous as well as discrete spectral lines.

Ans: (a) and (b): The Bohr model for the spectra of a hydrogen atom is applicable for only one electron systems. So, it cannot be applied to hydrogen in the molecular form and for helium atom.

12.11 The Balmer series for the H-atom can be observed

- (a) If we measure the frequencies of light emitted when an excited atom falls to the ground state.
- (b) If we measure the frequencies of light emitted due to transitions between excited states and the first excited state.
- (c) In any transition in a H-atom.
- (d) As a sequence of frequencies with the higher frequencies getting closely packed.

Ans: (b) and (d): The spectrum of light formed when the electron jumps from a higher energy level to the first energy level is called Balmer series. The spectral lines starts becoming close as the electron jumps from higher energy level to the ground level. Hence, options (b) and (d) are the correct answers.

12.12 Let $E_n = \frac{-1me^4}{8\varepsilon_0^2 n^2 h^2}$ be the energy of the nth level of H-atom. If all the H atoms are in the ground state and radiation of frequency $\frac{(E_2 - E_1)}{h}$ falls on it,

- (a) It will not be absorbed at all
- (b) Some of atoms will move to the first excited state.
- (c) All atoms will be excited to the state.
- (d) No atoms will make a transition to the $n = 3$ state.

Ans: (b) and (d): When the energy of radiation of photons is $\frac{(E_2 - E_1)n}{h}$, then electron jumps in next energy level ($n = 2$) after receiving this energy equal to the $E_2 - E_1$ energy. The new state is its unstable state. Electron jumps from E_2 to E_1 by radiating the energy of same frequency i.e., to $(E_2 - E_1)$. Electron can jump in next orbit. So Electron from ground state will jump at $n = 2$ not $n = 3$.

12.13 The simple Bohr model is not applicable to He_4 atom because

- (a) He_4 is an inert gas.

(b) He4 has neutrons in the nucleus.

(c) He4 has one more electron.

(d) Electrons are not subject to central forces.

Ans: (c) and (d): The Bohr's atomic model is applicable only for one electron systems and in there are two electrons in He_4 atom. In multi electron systems, electrons are not subject to central forces due to longer distance than nuclear size. Hence, options (c) and (d) are the correct answers.

Very Short Answer Questions

12.14 The mass of a H-atom is less than the sum of the masses of a proton and electron. Why is this?

Ans: Some energy is needed to bind the nucleons together in the nucleus in the process of formation of an atom with electrons and nucleons. This energy can be calculated from Einstein mass energy relation

$$E = \Delta m C^2$$

$$\text{Where } \Delta m = [Zm_p + (A - Z)m_n] - M$$

So the mass of an H atom is equal to

$$m_p + m_e - \frac{B.E.}{C^2}$$

Where B.E. represents the binding energy and is equal to 13.6eV .

12.15 Imagine removing one electron from ${}_2He^4$ and ${}_2He^3$. Their energy levels, as worked out on the basis of Bohr model will be very close. Explain why.

Ans: If we remove one electron from the isotopes of ${}_2He^4$ and ${}_2He^3$, both the isotopes will become one electron systems but the mass of the nucleus of the isotopes will be heavier than the hydrogen atom. So the energy levels as worked out on the basis of Bohr model will be very close.

12.16 When an electron falls from a higher energy to a lower energy level, the difference in the energies appears in the form of electromagnetic radiation. Why cannot it be emitted as other forms of energy?

Ans: When an electron falls from a higher energy level to a lower energy level, the charged particle experiences acceleration. The accelerated charged particle can produce electromagnetic wave only.

12.17 Would the Bohr formula for the H-atom remain unchanged if proton had a charge $\left(\frac{+4}{3}\right)e$ and electron a charge $\left(\frac{-3}{4}\right)e$, where $e = 1.6 \times 10^{-19} \text{ C}$. Give reasons for your answer?

Ans: The new charges on the proton and electron are given as $\left(\frac{+4}{3}\right)e$ and $\left(\frac{-3}{4}\right)e$ respectively. So, we find the product of the charges in the new case as the product of charges matters in the coulomb forces rather than the individual charges.

$$\frac{+4}{3}e \times \frac{-3}{4}e = -e^2$$

Originally, charge on electron is $-e$ and charge on proton is e .

Product is $-e^2$.

So, the product of charges is same as earlier.

As there is no change in position of protons and electrons and the product of charges is also same. Hence, the Bohr's formula for the new H-atom remain same.

12.18 Consider two different hydrogen atoms. The electron in each atom is in an excited state. Is it possible for the electrons to have different energies but the same orbital angular momentum according to the Bohr model?

Ans: In excited state of electrons of two H-atoms, electrons may be in orbit or energy level either $n = 2, 3, \dots$ and can have same energy but angular momentum by

Bohr's model is $L = \frac{nh}{2\pi}$. As n for both may be different so both H-atom will have different angular momentum.

Short Answer Questions

12.19 Positronium is just like a H-atom with the proton replaced by the positively charged anti-particle of the electron (called the positron which is as massive as the electron). What would be the ground state energy of positronium?

Ans: In the new specie, we have a positron in place of a proton in comparison to hydrogen atom.

So, mass of positronium $= m = m_e^- + m_e^+$

$$\text{Hence, } m_e^- = m_e^+ = \frac{m_p}{2}$$

As $E_n = -13.6$ and so energy of positron

$$E'_n = \frac{-m_e^+ e^4}{8\epsilon_0 n^2 h^2} = \frac{-\left(\frac{m_p}{2}\right) e^4}{8\epsilon_0 n^2 h^2} = \frac{-13.6}{2}$$

Therefore, we get, $E'_n = \frac{13.6}{2} \left(\because m_e = \frac{m_p}{2} \right)$

$$E'_n = -6.8 \text{ eV}$$

12.20 Assume that there is no repulsive force between the electrons in an atom but the force between positive and negative charges is given by Coulomb's law as usual. Under such circumstances, calculate the ground state energy of a He-atom.

Ans: For H atom $Z = 1$ and $n = 1$

$$\therefore E_n = \frac{-m_e e^4}{8\epsilon_0^2 1^2 h^2} = 13.6 \text{ eV}$$

For helium atom, $Z = 4$ and $n = 1$

$$\therefore E_n = \frac{-4m_e e^4}{8\epsilon_0^2 1^2 h^2} = -4 \times 13.6 = -54.4 \text{ eV} = -54.4 \text{ eV}$$

12.21 Using Bohr model, calculate the electric current created by the electron when the H-atom is in the ground state.

Ans: Let the velocity of electron of hydrogen atom in the orbit be v meters per second.

So, radius of orbit $= a_0$, where a_0 is the Bohr's radius.

So, the number of revolutions per second $= \frac{2\pi a_0}{v}$

Now, we know that $I = -ev$

Since we have, $v = \frac{2\pi a_0}{v}$

So, we get, $I = \frac{-e2\pi a_0}{v}$

(-) sign shows that the direction of current is opposite to the direction of motion of electron.

12.22 Show that the first few frequencies of light that is emitted when electrons fall to the n th level from levels higher than n , are approximate harmonics (i.e. in the ratio 1 : 2 : 3 ...) when $n \gg 1$.

Ans: When an electron falls from $(n + n')^{\text{th}}$ to n^{th} energy level, the frequency of radiations in spectrum Hydrogen atom like atoms is given by:

$$\nu = CRZ^2 \left[\frac{1}{(n + n')^2} - \frac{1}{n^2} \right]$$

In this case, $n' \gg n$

So, $n' = 1, 2, 3, \dots$ and R is the Rydberg's constant. So, we get,

$$\nu = CRZ^2 \left[\frac{1}{n^2 \left[1 + \frac{n'}{n^2} \right]^2} - \frac{1}{n^2} \right] = CRZ^2 \left[\frac{1}{n^2} \left[1 + \frac{n'}{n} \right]^{-2} - \frac{1}{n^2} \right]$$

Neglecting the higher terms as $n \gg n'$,

$$\begin{aligned} &= CRZ^2 \left[\frac{1}{n^2} \left(1 - \frac{2n'}{n} \right) - \frac{1}{n^2} \right] = CRZ^2 \left[\frac{1}{n^2} - \frac{2n'}{n^3} - \frac{1}{n^2} \right] \\ &= \frac{-CRZ^2 2n'}{n^3} = \left(\frac{2CRZ^2}{n^3} \right) n' \end{aligned}$$

So the first few frequencies of light that is emitted when electrons fall from $(n + n')$ to n^{th} energy level are in the ratio of $n' = 1, 2, 3, \dots$ when $n \gg 1$.

12.23 What is the minimum energy that must be given to a H atom in ground state so that it can emit an H_γ line in Balmer series? If the angular momentum of the system is conserved, what would be the angular momentum of such H_γ photon?

Ans: we know that H spectral line in Balmer series formed when electron falls from $n = 5$ to $n = 1$. Here the electron is in ground state i.e., $n = 1$ and must be taken to $n = 5$ for H line. So the energy of

$$H? = E_5 - E_1 = \left(\frac{13.6}{5^2} \right) - (-13.6) = -0.54 + 13.6 = 13.06 \text{ eV}$$

Since angular momentum is conserved, so the angular momentum of

$H\gamma$ = Change in angular momentum of electron

$$= L_5 - L_1 = 5h - 2h$$

$$= 3h = 3 \times 6.63 \times 10^{-34} = 19.89 \times 10^{-34} \text{ kgm}^2/\text{s}$$

Long Answer Type Questions

12.24 The first four spectral lines in the Lyman series of a H-atom are $\lambda = 1218 \text{ \AA}$, $1028 \text{ \AA}$, $974.3 \text{ \AA}$ and $951.4 \text{ \AA}$. If instead of Hydrogen, we consider Deuterium, calculate the shift in the wavelength of these lines.

Ans: Reduced mass of H atom (mass defect) μ_H then

$$\frac{1}{\mu_H} = \frac{1}{m_e} + \frac{1}{M} \quad (M \text{ is mass of H atom})$$

$$\frac{1}{\mu_H} = \frac{M + m_e}{M \cdot m_e} = \frac{M \left[1 + \frac{m_e}{M} \right]}{M \cdot m_e} \quad (\because M \gg m_e)$$

$$\therefore \mu_H = m_e \left[1 + \frac{m_e}{M} \right]^{-1} = m_e \left[1 - \frac{m_e}{M} \right]$$

For Deuterium $M = 2M$

Reduced mass of Deuterium $= \mu_D$

$$\mu_D = m_e \left[1 - \frac{m_e}{M} \right] \left[1 + \frac{m_e}{M} \right]$$

Total energy E_n of the electron revolving in an n th stationary orbit,

$$E_n = \frac{-me^4}{8n^2 g e_0^2 h^2}$$

m is the reduced mass of the electron and proton in H atom.

So $h\nu = E_{n_i} - E_{n_f}$

$$\nu = \frac{me^4}{8\varepsilon_0^2 h^2} \left[\frac{1}{n_i^2} - \frac{1}{n_f^2} \right] = \frac{c}{\lambda}$$

$\nu \propto m$ (Reduced mass)

$$\frac{1}{\lambda} \propto \mu \Rightarrow \lambda \propto \frac{1}{\mu}$$

For hydrogen and deuterium $= \frac{\lambda_H}{\lambda_D}$

$$\text{So } \frac{\lambda_H}{\lambda_D} = \frac{\mu_H}{\mu_D} = \frac{m_e \left[1 - \frac{m_e}{M} \right]}{m_e \left[1 - \frac{m_e}{M} \right] \left[1 + \frac{m_e}{2M} \right]}$$

$$\lambda_D = \left[1 + \frac{m_e}{2M} \right]^{-1} \lambda_H = \left(1 - \frac{m_e}{M} \right) \lambda_H$$

$$\lambda_D = \lambda_H (0.99973)$$

$$\lambda_H = \text{\AA}, \text{\AA}, \text{\AA} \text{ and } \text{\AA} \text{ (Given)}$$

$$\lambda_{D1} = 1214 \text{\AA}$$

$$\lambda_{D2} = 0.9973 \times 1028 = 1025 \text{\AA}$$

$$\lambda_{D3} = 0.9973 \times 974 = \text{\AA}$$

$$\lambda_{D4} = 0.9973 \times 954 = \text{\AA}$$

12.25 Deuterium was discovered in 1932 by Harold Urey by measuring the small change in wavelength for a particular transition in $^1\text{H}_1$ and $^2\text{H}_2$. This is because, the wavelength of transition depend to a certain extent on the nuclear mass. If nuclear motion is taken into account then the electrons and nucleus revolve around their common center of mass. Such a system is equivalent to a single particle with a reduced mass μ , revolving around the nucleus at a

distance equal to the electron-nucleus separation. Here $\mu = \frac{m_e M}{(M + m)}$ where M

is the nuclear mass and m_e is the electronic mass. Estimate the percentage difference in wavelength for the 1st line of the Lyman series in $^1\text{H}_1$ and $^2\text{H}_2$. (Mass of $^1\text{H}_1$ nucleus is $1.6725 \times 10^{-27} \text{ kg}$, Mass of $^2\text{H}_2$ nucleus is $3.3374 \times 10^{-27} \text{ kg}$, Mass of electron = $9.109 \times 10^{-31} \text{ kg}$.)

Ans: Total energy of electron in nth stable orbit in H or like atom

$$E_n = \frac{\mu Z^2 e^4}{8 \epsilon_0^2 h^2 n^2}$$

μ = Reduced mass of electron, proton and neutron (mass defect)

$$E_H = \frac{\mu_H (1)^2 e^4}{8\epsilon_0^2 h^2} \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right] = \frac{\mu_H e^4}{8\epsilon_0^2 h^2} \left[\frac{1}{1} - \frac{1}{2^2} \right] = \frac{\mu_H e^4}{8\epsilon_0^2 h^2} \left[\frac{3}{4} \right]$$

$$E = h\nu = \frac{h}{\lambda} \text{ or } \lambda_H = \frac{h}{E_H}$$

$$\therefore h\nu_H = \frac{\mu_H e^4 \cdot \frac{3}{4}}{8\epsilon_0^2 h^2 \cdot 2}$$

$$\nu_H = \frac{\mu_H e^4}{8\epsilon_0^2 h^3} \cdot \frac{3}{4}$$

$$= \frac{(\lambda_D - \lambda_H)}{\lambda_H} \times 100 \text{ Percent change in wavelength}$$

Percent change in wavelength

$$\% \text{ change } \Delta E = \left[\frac{\lambda_D}{\lambda_H} - 1 \right] \times 100 \because \Delta E = E_1 - E_2 \dots\dots (I)$$

$$h\nu = \frac{\mu e^4}{8\epsilon_0^2 h^2} \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right]$$

$$\nu = \frac{\mu e^4}{8\epsilon_0^2 h^2} \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right]$$

$$\frac{c}{\lambda} = \frac{\mu e^4}{8\epsilon_0^2 h^2} \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right]$$

$$\frac{1}{\lambda} = \frac{\mu e^4}{8\epsilon_0^2 ch^2} \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right]$$

as μ = mass defect, e, ϵ_0 , and h are constants for an atom.

$$\therefore \lambda \propto \frac{1}{h}$$

So eqn. Ist can be written as percentage change in the wavelength = $\left[\frac{\mu_H}{\mu_D} - 1 \right] \times 100$

$$\therefore \mu = \frac{m_e M}{(M + m_e)} \text{ (Given)}$$

$$\therefore \text{Percentage change in wavelength} = \left[\frac{\frac{m_e M_H}{(M_H + m_e)}}{\frac{m_e + M_D}{(M_D + m_f)}} - 1 \right] \times 100$$

$$\frac{\Delta \lambda}{\lambda_H} \times 100 = \left[\frac{M_H}{M_D} \frac{(M_D + m_e)}{(M_H + m_f)} - 1 \right] \times 100$$

$$= \left[\frac{M_H}{M_D} \frac{M_D \left(1 + \frac{m_e}{M_D} \right)}{M_H \left(1 + \frac{m_e}{M_H} \right)} - 1 \right] \times 100$$

$$= \left[\left(1 + \frac{m_e}{M_D} \right) \left(1 + \frac{m_e}{M_H} \right)^{-1} - 1 \right] \times 100$$

$$= \left[\left(1 + \frac{m_e}{M_D} \right) \left(1 - \frac{m_e}{M_H} \right) - 1 \right] \times 100$$

$m_e \ll M_D$ So neglecting the higher degree term

$$\frac{\Delta \lambda}{\lambda_H} \times 100 = \left[1 - \frac{m_e}{M_H} + \frac{m_e}{M_D} - \frac{m_e m_e}{M_D - M_H} - 1 \right] \times 100$$

$$= m_e \left[\frac{1}{M_D} - \frac{1}{M_H} \right] \times 100$$

$$= 9.1 \times 10^{-31} \left[\frac{1}{3.3374 \times 10^{-27}} - \frac{1}{1.6725 \times 10^{-27}} \right] \times 100$$

$$= \frac{9.1 \times 10^{-31+2}}{10^{-27}} \left[\frac{1.6725 - 3.3374}{3.3374 \times 1.6725} \right]$$

$$\frac{\Delta \lambda \times 100}{\lambda_H} = \frac{-9.1 \times 10^{-20+27} \times 0.6649}{3.3374 \times 1.6725} = \frac{-6.05059 \times 10^{-2}}{5.5180}$$

$$\frac{\Delta \lambda \times 100}{\lambda_H} = -1.084 \times 10^{-2} \% \text{ Decrease in wavelength.}$$

(-) sign shows that $\lambda_D < \lambda_H$

12.26 If a proton had a radius R and the charge was uniformly distributed, calculate using Bohr theory, the ground state energy of a H-atom when (i) $R = 0.1\text{\AA}$, and (ii) $R = 10\text{\AA}$.

Ans: (i) Consider in H atom nucleus as point charge electron is revolving around nucleus with speed v and radius r_A . The Colombian force provides centripetal force to revolve around nucleus.

$$\therefore \frac{m_e v^2}{r_A} = \frac{-Ke^2}{r_A^2} \dots (I)$$

$$\text{Here } K = \frac{1}{4\pi\epsilon_0}$$

(-) sign shows the force of attraction.

By Bohr's postulate, angular momentum = $\frac{nh}{2\pi}$

$$mvr_A = \frac{nh}{2\pi}$$

$$v = \frac{nh}{2\pi mr_A}$$

$$\frac{mn^2h^2}{4\pi^2 m^2 r_A^2 r_A} = \frac{Ke^2}{r_A^2} [\text{From I}]$$

$$r_A = \frac{n^2 h^2}{4\pi^2 m K e^2} \dots (II)$$

For ground state $n = 1$

$$r_A = \frac{h^2}{4\pi^2 m K e^2}$$

$$\begin{aligned} &= \frac{6.63 \times 10^{-34} \times 6.63 \times 10^{-34}}{(2 \times 3.14)^2 \times 9.1 \times 10^{-31} \times 9 \times 10^9 \times 1.6 \times 10^{-19} \times 1.6 \times 10^{-19}} \\ &= \frac{6.63 \times 6.63 \times 10^{-68+38+31-9}}{9.19 \times 1.6 \times 1.6 \times 4 \times 3.14 \times 4 \times 3.14} \end{aligned}$$

$$= r_A = 0.53 \text{ \AA}$$

$$P.E. = \frac{-Ke^2}{r_A} = \frac{-9 \times 10^9 \times 1.6 \times 10^{-19} \times 1.6 \times 10^{-19}}{0.53 \times 10^{-10}} J$$

$$= \frac{-9 \times 1.6 \times 1.6 \times 10^{-19} \times 10^{-19} \times 10^9}{0.53 \times 10^{-10} \times 1.6 \times 10^{-19}} J$$

$$P.E. = \frac{14.4}{0.53} = 27.17 = 27.2 \text{ eV}$$

$$K.E = \frac{1}{2} mv^2$$

$$= \frac{1}{2} \frac{m \cdot n^2 h^2}{4\pi^2 m^2 r^2} = \frac{1}{2} \frac{h^2}{4\pi^2 m r^2} \text{ (n = 1 for ground state)}$$

$$= \frac{1}{2} \frac{6.62 \times 10^{-34} \times 6.62 \times 10^{-34}}{4 \times 3.14 \times 3.14 \times 9 \times 10^{-31} \times 0.53 \times 10^{-10} \times 0.53 \times 10^{-10}} J$$

$$= \frac{6.62 \times 6.62 \times 10^{-68+31+10+10}}{4 \times 3.14 \times 3.14 \times 18 \times 0.53 \times 0.53 \times 1.6 \times 10^{-19}} eV$$

$$= \frac{6.62 \times 6.62 \times 10^{-68+51+19}}{4 \times 3.14 \times 3.14 \times 18 \times 0.53 \times 0.53}$$

$$K.E. = 0.1373 \times 10^2 \text{ eV} = 13.7 \text{ eV}$$

$$P.E. = 27.2 \text{ eV}$$

(ii) Now for spherical nucleus of radius, R, electron moves charge inside the nucleus

$R \gg r_b$ then electron moves inside the nucleus. Then (r_b is radius of new Bohr's orbit of revolving electron)

Charge

$$e = \frac{e \cdot \left(\frac{4}{3} \pi r_b^3 \right)}{\frac{4}{3} \pi R^3}$$

$$e' = q_2 = \frac{e r_b^3}{R^3}$$

$$q_1 = e$$

$$\frac{mv^2}{r_b} = \frac{Kee'}{r_b^2} \text{ (By Coulomb's law)}$$

$$mv_b = \frac{nh}{2\pi} \Rightarrow v_b = \frac{nh}{2\pi mr_b} \text{ (By Bohr's postulate)}$$

$$\therefore \frac{m}{r_b} \frac{n^2 h^2}{4\pi^2 m^2 r_b^2} = \frac{kee^1}{r_b^2}$$

$$r_b = \frac{n^2 h^2}{4\pi^2 m kee'}$$

Now for ground state of H, $n=1$ and $e' = \frac{er_A^3}{R^3}$, then

$$\therefore r_b = \frac{h^2}{4\pi^2 m K \cdot e \cdot e \cdot \frac{r_b^3}{R^3}} = \left(\frac{h^2}{4\pi^2 m K e^2} \right) \times \frac{R^3}{r_b^3} = r_A \times \frac{R^3}{r_b^3}$$

$$[r_A = \frac{h^2}{4\pi^2 m K e^2} = 0.53 \text{ \AA} \text{ Calculated in part (i)}]$$

$$r_b = r_A \left(\frac{R}{r_b} \right)^3$$

$$r_b^4 = r_A R^3 = 0.53 \text{ \AA} \times (10 \text{ \AA})^3$$

$$r_b^4 = 0.53 \times 1000 (\text{ \AA})^4 (\because r_A = 0.53 \text{ \AA})$$

$$r_b = [530 \text{ \AA}^4]^{1/4} \text{ \AA} < R \text{ \AA}$$

$$\begin{aligned} K.E. &= \frac{1}{2} mv^2 = \frac{m}{2} \cdot \frac{h^2}{4\pi^2 m^2 r_b^2} = \frac{h^2}{8\pi^2 m r_b^2} \left[\because v = \frac{n^2 h^2}{4\pi^2 m^2 r_b^2} \right] \\ &= \frac{6.62 \times 6.62 \times 10^{-34} \times 10^{-34}}{8 \times 3.14 \times 3.14 \times 9.1 \times 10^{-31} \times 4.8 \times 4.8 \times 10^{-20} \times 1.6 \times 10^{-19}} \text{ eV} \\ &= \frac{6.62 \times 6.62 \times 10^{-68+31+20+19}}{8 \times 3.14 \times 3.14 \times 9.1 \times 4.8 \times 4.8 \times 1.6} \text{ eV} \\ &= \frac{43.8244}{26460.2} 10^{-68+70} = 0.001656 \times 10^2 \text{ eV} \end{aligned}$$

$$K.E = 0.167 \text{ eV}$$

$$P.E. = \frac{e^2}{4\pi\epsilon_0} \frac{(r_b^2 - 3R^2)}{R^3} \left(P.E. = \frac{Kq_1q_2}{r} \right)$$

$$P.E. = \left[\frac{e^2}{4\pi\epsilon_0 r_A} \right] \frac{r_A(r_b^2 - 3R^2)}{R^3} \left[\text{multiplying by } \frac{r_A}{r_A} \right]$$

From part (i)

$$P.E. = \frac{e^2}{4\pi\epsilon_0 r_A} = 27.2 \text{ eV}$$

$$\therefore P.E. = 27.2 \left[\frac{0.53(\sqrt{530} - 300)}{1000} \right] \left[\because r_b = (530)^{1/4} \text{ \AA} \text{ and } R = 10 \text{ \AA} \right]$$

$$= \frac{27.2 \text{ eV} \times 0.53(23.02 - 300) \text{ \AA}^3}{1000 \text{ \AA}^3} = 27.2 \times \frac{0.53(-276.9)}{1000} \text{ eV}$$

$$P.E. = \frac{3992.9}{1000} = -3.99 \text{ eV}$$

$$K.E. = 0.167 \text{ eV}$$

12.27 In the Auger process an atom makes a transition to a lower state without emitting a photon. The excess energy is transferred to an outer electron which may be ejected by the atom. (This is called an Auger electron). Assuming the nucleus to be massive, calculate the kinetic energy of an $n = 4$ Auger electron emitted by Chromium by absorbing the energy from a $n = 2$ to $n = 1$ transition.

Ans: The energy E_n of the n th state

$$E_n = +Z^2 R \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right] = Z^2 R \left(\frac{1}{1} - \frac{1}{4} \right) \left(\text{for } n_1 = 1, n_2 = 2 \right)$$

$$Z = 24$$

R = Rydberg constt.

$$\therefore E_n = \frac{3}{4} Z^2 R$$

The energy required to eject an electron from $n = 4$ state is

$$E_4 = Z^2 R \frac{1}{4^2} = \frac{1}{16} Z^2 R$$

Energy given to electron is converted into K.E. of ejected electron.

Hence, the K.E. of Auger (ejected) electron = $E_n - E_4$

$$K.E. = Z^2 R \frac{3}{4} - \frac{1}{16} Z^2 R = \frac{11}{16} Z^2 R = \frac{11}{16} \times 24 \times 24 \times 13.6 \text{ eV}$$

$$K.E. = 11 \times 36 \times 13.6 = 5385.6 \text{ eV}$$

12.28 The inverse square law in electrostatics is $|F| = \frac{e^2}{(4\pi\epsilon_0)r^2}$ for the force

between an electron and a proton. The $\frac{1}{r}$ dependence of $|F|$ can be understood in quantum theory as being due to the fact that the 'particle' of light (photon) is massless. If photons had a mass m_p , force would be modified

to $\left[F = \frac{e^2}{(4\pi\epsilon_0)r^2} \left[\frac{1}{r^2} + \frac{\lambda}{r} \right] e^{-\lambda r} \right]$ where $\lambda = \frac{m_p c}{\hbar}$ and $\hbar = \frac{h}{2\pi}$

Estimate the change in the ground state energy of a H-atom if m_p were 10^6 times the mass of an electron.

Ans: Mass of photon = $9.1 \times 10^{-31} \times 10^6 \text{ kg}$

$$= 9.1 \times 10^{-37} \text{ kg}$$

Wavelength associated with a photon = $\frac{h}{m_p c}$

$$\lambda = \frac{6.62 \times 10^{-34}}{9.1 \times 10^{-37} \times 3 \times 10^8}$$

$$= \frac{6.62}{9.1 \times 3} \times 10^{-34+37-8} = 2.4 \times 10^{-7} \gg r_A \text{ (see Q.26)}$$

$$\lambda \ll \frac{1}{r_A} < e\hat{A}_A \ll 1$$

$$U(r) = \frac{-e^2 e^{-\lambda r}}{4\pi\epsilon_0 r}$$

$$mvr = \frac{h}{2\pi} = \hbar_{or} v = \frac{\hbar}{mr} \dots (I)$$

$$\frac{mv^2}{v} \frac{e^2}{4\pi\epsilon_0} \left[\frac{1}{r^2} + \frac{\lambda}{r} \right] \left[\because F = \frac{e^2}{(4\pi\epsilon_0)r^2} \left[\frac{1}{r^2} + \frac{\lambda}{r} \right] e^{-ir} \text{ given} \right]$$

$$\frac{m}{r} \cdot \frac{\hbar^2}{m^2 r^2} = \frac{e^2}{4\pi\epsilon_0} \left[\frac{1+\lambda r}{r^2} \right]$$

$$\frac{\hbar^2}{mv} = \frac{e^2}{4\pi\epsilon_0} (1+\lambda r)$$

$$\frac{\hbar^2 4\pi\epsilon_0}{me^2} = (r + \lambda r^2)$$

$$\text{If } \lambda = 0, \frac{\hbar^2 4\pi\epsilon_0}{m_0} [\text{neglecting } r^2]$$

$$\frac{\hbar}{m} = \frac{e^2}{4\pi\epsilon_0} r_A (r_A = r + \lambda r^2)$$

$$\because \lambda_A \gg r_B \text{ And } \delta$$

$$\text{Taking } r_A + r + \lambda r^2 [\because \delta]$$

$$r_A = (r_A \delta) + M(r_A \delta) \text{ (put } S^2)$$

$$0 = \delta + \lambda r_A^2 + 2r_A \delta \lambda \text{ (Neglecting small term } \delta^2)$$

$$0 = \delta + 2r_A \delta \lambda + \lambda r_A^2$$

$$\Rightarrow \delta [1 + 2r_A \lambda] = -\lambda r_A^2$$

$$\delta = \frac{-\lambda r_A^2}{(1 + 2r_A \lambda)} = -\lambda r_A^2 (1 + 2r_A \lambda)^{-1}$$

$$\delta = -\lambda r_A^2 [1 - 2r_A \lambda] = -\lambda r_A^2 + 2r_A^3 \lambda^2$$

$$\therefore \lambda \text{ And } r_A \ll 1 \text{ so } r_A^3 \lambda^2 \text{ is very small so by neglecting it we get,}$$

$$\delta = -\lambda r_A^2$$

$$\begin{aligned}
 V(r) &= \frac{-e^2}{4\pi\epsilon_0} = \frac{e^{(-2\delta - \lambda r_A)}}{(r_A + \delta)} = \frac{-e^2}{4\pi\epsilon_0} \cdot \frac{e^{-2(\delta + r)}}{r_A \left(1 + \frac{\delta}{r_A}\right)} \\
 &= \frac{-e^2}{4\pi\epsilon_0 r_A} e^{-\lambda r} \left(1 + \frac{\delta}{r_A}\right)^{-1} \because \left(\frac{1}{1+x}\right)^{-1} \\
 &= \frac{-e^2 e^{-\lambda r}}{4\pi\epsilon_0 r_A} \left(1 - \frac{\delta}{r_A}\right)
 \end{aligned}$$

$V(r) = -27.2\text{eV}$ Remains unchanged

$$\begin{aligned}
 K.E. &= \frac{1}{2}mv^2 = \frac{1}{2}m\left(\frac{h}{mr}\right)^2 = \frac{1}{2}\frac{h^2}{mr^2} \left(\text{From } I, v = \frac{h}{mr}\right) \\
 &= \frac{-\hbar^2}{2m(r_A + \delta)^2} = \frac{-\hbar^2}{2mr_A^2\left(1 + \frac{\delta}{r_A}\right)^2} = \frac{\hbar^2}{2mr_A^2}\left(1 - \frac{2\delta}{r_A}\right) \\
 &= \frac{\hbar^2}{2mr_A}\left(1 - \frac{\lambda r_A^2}{r_A}\right) = \frac{\hbar^2}{2mr_A}(1 + 2\lambda r_A) \\
 &= 13.6\text{eV}(1 + 2\lambda r_A)
 \end{aligned}$$

$$\begin{aligned}
 \text{Total energy} &= \frac{-e^2}{4\pi\epsilon_0 r_A} + \frac{\hbar^2}{2mr_A^2}(1 + 2\lambda r_A) \\
 &= [-27.2 + 13.6(1 + 2\lambda r_A)]\text{eV} \\
 &= -27.2 + 13.6 + 27.2\lambda r_A\text{eV}
 \end{aligned}$$

$$\text{Total } E = -13.6 + 27.2\lambda r_A$$

$$\text{Change in energy} = -13.6 + 27.2\lambda r_A - (-13.6) = 27.2\lambda r_A\text{eV}$$

12.29 The Bohr model for the H-atom relies on the Coulomb's law of electrostatics. Coulomb's law has not directly been verified for very short distances of the order of angstroms. Supposing Coulomb's law between two opposite's charges $+q_1, -q_2$ is modified to

$$|F| = \frac{q_1 q_2}{4\pi\epsilon_0 r^2} (r \geq R_0)$$

$$|F| = \frac{q_1 q_2}{4\pi\epsilon_0 R_0^2} \left[\frac{R_0}{r} \right]^z (r \leq R_0)$$

Calculate in such a case, the ground state energy of a H-atom, if $\epsilon = 0.1$ and $R_0 = 1\text{\AA}$.

Ans: Case I: when $r \leq R_0 = 1\text{\AA}$

Let $\epsilon = 2 + \delta$

$$F = \frac{q_1 q_2}{4\pi\epsilon_0 R_0^2} \frac{R_0^{r+\delta}}{r^{2+\delta}} = \left(\frac{e \cdot (-e)}{4\pi\epsilon_0} \right) \frac{R_0^\delta}{r^{2+\delta}} [q_1 = e, q_2 = -e]$$

$$|F| = -\left(1.6 \times 10^{-18}\right)^2 \times 9 \times 10^9 \frac{R_0^\delta}{r^{2+\delta}} [(-)] [(-)\text{ve sign shows force of attraction}]$$

$$|F| = 23.04 \times 10^{-29} \frac{R_0^\delta}{r^{2+\delta}}$$

The electrostatic force of attraction between positively charged nucleus and negatively charged electron provides necessary centripetal force.

$$\frac{mv^2}{r} = \left(23.04 \times 10^{-29}\right) \frac{R_0^\delta}{r^{2+\delta}}$$

$$\text{Let } 23.04 \times 10^{-29} = \Lambda$$

$$\therefore v^2 = \Lambda \frac{R_0^\delta}{r^{2+\delta}} \cdot \frac{r}{m} = \frac{\Lambda R_0^\delta}{mr^{1+\delta}}$$

By Bohr's IInd postulate, angular momentum

$$L = \frac{nh}{2\pi} \text{ and } \hbar = \frac{h}{2\pi}$$

$$\therefore L = nh$$

$$mvr = nh$$

$$r = \frac{nh}{mv} = \frac{nh}{m} \sqrt{\frac{mr^{1+\delta}}{\Lambda R_0^\delta}} = \frac{nh}{m} \left[\frac{[m]}{\Lambda R_0^\delta} \right]^{\frac{1}{2}} r^{1/2+\delta/2}$$

$$r^{1-\frac{1}{2}-\frac{\delta}{2}} = \frac{n\hbar}{m} \left[\frac{m}{\Lambda R_0^\delta} \right]^{1/2}$$

$$r^{\frac{1}{2}(1-\delta)} = \frac{n\hbar}{m} \left[\frac{m}{\Lambda R_0^\delta} \right]^{-1/2}$$

$$\gamma_n = \left[\frac{n^2 \hbar^2}{m \Lambda R_0^\delta} \right]^{\frac{1}{1-\delta}}$$

$$= \left[\frac{\hbar^2}{m \Lambda R_0^\delta} \right]^{\frac{1}{1-\delta}} \quad (n=1 \text{ for ground state})$$

$$r_1 = \left[\frac{(1.05 \times 10^{-34})^2}{9.1 \times 10^{-31} \times 23.04 \times 10^{-29} \times 10^{19}} \right]^{\frac{1}{29}}$$

$$\left[\because \hbar = \frac{h}{2\pi} = \frac{6.6 \times 10^{-34}}{2 \times 3.14} = (1.05 \times 10^{-34}) \right]$$

Where $h = 1.05 \times 10^{-34} \text{ JS}^{-1}$

$R_0 = 10^{19}$ and $1 - \delta = 2.9$

$R_1 = 8 \times 10^{-11} \text{ m} = 0.08 \text{ nm}$

This is a radius of orbit of electron in ground state of hydrogen atom

Velocity of electron in ground state By IInd postulate of Bohr's Atomic model

$$mv_n r_n = \frac{n\hbar}{2\pi} \left(\hbar = \frac{h}{2\pi} \right)$$

$$v_n = \frac{n\hbar}{mr_n} = \frac{n\hbar}{m} \left[\frac{m \Lambda R_0^b}{n^2 \hbar^2} \right]^{\frac{1}{1-\delta}}$$

$$\text{Case II: } n=1, v_1 = \frac{\hbar}{mr_1} = \frac{1.05 \times 10^{-34}}{9.1 \times 10^{-31} \times 0.08 \times 10^{-9}} = \frac{1.05}{0.728} \times 10^{-34+31+9}$$

$$= 1.44 \times 10^{-34+40} = 1.44 \times 10^6 \text{ m/s}$$

$$v_1 = 1.44 \times 10^6 \text{ m/s}$$

$$\begin{aligned}
 KE &= \frac{1}{2}mv_1^2 - 9.43 \times 10^{-19} \\
 &= \frac{1}{2} \times 9.1 \times 10^{-31} \times (1.44 \times 10^5)^2 - 9.43 \times 10^{-19} \text{ J} \\
 &= 5.9 \text{ eV}
 \end{aligned}$$

P.E. from R_0 to $\frac{-\Lambda}{R_0}$

P.E from R_0 to $r = +\Lambda R_0^\theta \int_{R_0}^r \frac{dr}{r^{2+\delta}}$

This is the P.E. of electron in ground state R_0 to $r = \frac{-\Lambda}{R_0}$

$$\begin{aligned}
 P.E. &= -\frac{\Lambda R_0^\delta}{(1+\delta)} \left[\frac{1}{r^{1+\delta}} - \frac{1}{R_0^{1+\delta}} \right] \\
 &= \frac{-\Lambda}{(1+\delta)} \left[\frac{R_0^\delta}{r^{1+\delta}} - \frac{1}{R_0} \right] = \frac{-\Lambda}{(1+\delta)} \left[\frac{R_0^\delta}{r^{1+\delta}} - \frac{1}{R_0} + \frac{1+\delta}{R_0} \right]
 \end{aligned}$$

Put $\delta = -1.9$

$$\begin{aligned}
 P.E. &= \frac{-\Lambda}{(1+(-1.9))} \left[\frac{R_0^{-1.9}}{r^{-0.9}} - \frac{1}{R_0} - \frac{1.9}{R_0} \right] \\
 &= \frac{-\Lambda}{-0.9} \left[\frac{R_0^{-1.9}}{r^{-0.9}} - \frac{1.9}{R_0} \right] = \frac{23.04 \times 10^{-29}}{0.9} [(0.8)^{0.9} - 1.9] \text{ J} \\
 &= 17.3 \text{ eV}
 \end{aligned}$$

Total E = P.E. + K.E.

$$= (-17.3 + 5.9) \text{ eV}$$

Total energy = -11.4 eV

This is the required total energy of electron in ground state of H-atom.